

E-COMMERCE SALES ANALYSIS USING SQL

1.0 PROJECT OVERVIEW

This project used **SQL** to analyze an e-commerce sales dataset. The analysis looked at customers, products, employees, orders, sales, countries and discounts.

The main aim was to find useful information from the data and use it to make better business decisions.

Tools Used: MySQL and MySQL Workbench

2.0 PROJECT OBJECTIVES

The analysis was carried out to:

1. Find the employees with the highest sales.
2. Identify the customers who spent the most.
3. Check how sales changed over time.
4. Understand how customers buy products.
5. Find products with high and low sales.
6. Compare sales across countries and product categories.
7. Examine the effect of discounts on orders.
8. Find useful business insights and recommendations.

3.0 ANALYSIS AND FINDINGS

3.1 Employee Sales Performance

Finding: Margaret Peacock had the highest sales at \$9,642, followed by Nancy Davolio with \$5,204.

What this means: Margaret Peacock made more sales than the other employees in the analysis.

3.2 Highest-Spending Customers

Finding: Ernst Handel was the highest-spending customer among the top five, with \$3,642 in total purchases.

What this means: Some customers are more valuable to the business because they spend more.

3.3 Monthly Sales

Finding: Sales were \$6,226 in January 1997 and \$2,091 in February 1997.

What this means: The business made much more sales in January than in February.

3.4 Order Fulfilment Time

Finding: The average fulfilment time ranged from 3.00 to 3.84 days.

What this means: Most fulfilment times were fairly close, although some were slightly longer.

3.5 Top Customer in London

Finding: Seven Seas Imports recorded the highest sales at \$1,064.

What this means: Seven Seas Imports was an important customer in the London market.

3.6 Customers Who Ordered More Than Once in a Day

Finding: Bottom-Dollar Markets placed 2 orders on January 10, 1997.

What this means: The customer made more than one purchase on the same day.

3.7 Product Discounts

Finding: The average discounts for the selected products were between 7% and 9%.

What this means: Different products were sold with different discount levels.

3.8 Customer Buying Behaviour

Finding: Some customers placed very large orders. Save-a-lot Markets ordered 120 units of Pâté chinois, while QUICK-Stop ordered 100 units of selected products.

What this means: Some customers have much higher demand for certain products.

3.9 Employee Sales Ranking

Finding: Margaret Peacock ranked 1st, followed by Nancy Davolio and Janet Leverling.

What this means: Margaret Peacock had the highest total sales among the employees analyzed.

3.10 Sales by Country and Product Category

Finding: Sales were different across countries and product categories.

What this means: Some countries and product categories performed better than others.

3.11 Product Sales Growth

Finding: Gustaf's Knäckebröd recorded a 700% increase, while Inlagd Sill recorded a 99.26% decrease.

What this means: Some products became much more popular, while others lost most of their sales.

3.12 Order Size

Finding: The dataset contained both small and large orders.

What this means: Customers do not all buy the same amount.

3.13 Products with Very Few Orders

Finding: Mishi Kobe Niku, Røgede sild, Chocolate and Laughing Lumberjack Lager appeared in only one order.

What this means: These products had very low demand in the available data.

3.14 Highest-Revenue Product

Finding: Chang generated the highest identified revenue in the Beverages category at \$1,102.

What this means: Chang was the best-performing product in the selected category based on revenue.

3.15 High-Value Discounted Orders

Finding: Several orders met these conditions, including Order 10252 worth \$395 and Order 10258 worth \$233.

What this means: Some customers placed large orders while receiving discounts.

BUSINESS RECOMMENDATIONS

Based on the analysis, the company should:

1. Reward and learn from top employees to improve overall sales performance.
2. Focus on high-spending customers and use rewards to encourage them to keep buying.
3. Study high-sales periods and use successful strategies during slower periods.
4. Keep popular products in stock to avoid missing sales opportunities.
5. Review products with low sales and decide whether they need better pricing, promotion, or reduced stock.
6. Check discounts carefully to make sure they increase sales without reducing profit too much.
7. Pay attention to high-volume customers and provide good service to maintain their business.
8. Focus on strong countries and product categories while finding ways to improve weaker areas.

SQL SKILLS USED

The project used the following SQL skills:

- SELECT
- WHERE
- JOIN
- GROUP BY
- HAVING
- Aggregate functions
- CASE statements
- Date functions
- Subqueries
- CTEs
- EXISTS
- RANK()
- LAG()
- PERCENT_RANK()
- Sorting and filtering
- Data analysis

CONCLUSION

This project shows how SQL can be used to turn sales data into simple and useful business information. The analysis helped identify top employees, valuable customers, popular and low-demand products, sales changes, customer buying patterns and the use of discounts.